

Hoja de fórmulas

1. $X \sim \text{Binomial}(n, p)$, $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$, $E(X) = np$, $V(X) = np(1-p)$
2. $X \sim \text{Geométrica}(p)$, $p_X(k) = p(1-p)^{k-1}$, $E(X) = \frac{1}{p}$, $V(X) = \frac{1-p}{p^2}$
3. $X \sim \text{Binomial Negativa}(r, p)$, $p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$, $E(X) = \frac{r}{p}$, $V(X) = \frac{r(1-p)}{p^2}$
4. $X \sim \text{Poisson}(\lambda)$, $p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $E(X) = \lambda$, $V(X) = \lambda$
5. $X \sim \text{Uniforme}(a, b)$, $f_X(k) = \begin{cases} \frac{1}{b-a} & \text{si } k \in (a, b) \\ 0 & \text{si } k \notin (a, b) \end{cases}$ $E(X) = \frac{a+b}{2}$, $V(X) = \frac{(b-a)^2}{12}$
6. $X \sim \text{Normal}(\mu, \sigma^2)$, $f_X(k) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(k-\mu)^2}{2\sigma^2}}$, $E(X) = \mu$, $V(X) = \sigma^2$

TEOREMA

Si $X_1, \dots, X_n$ son i.i.d. con $X_i \sim \text{Normal}(\mu, \sigma^2)$ entonces

- $\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim \text{Normal}(0, 1)$
- $\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim T_{n-1}$
- $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$

donde $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ y $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.